ESP32 ACCESS CONTROL SYSTEM

Project Task Allocation

Function inventory, module breakdown and member assignment for the Smart Building Access & Room Automation firmware.

PART 01

Code Function Inventory

26 functions across drivers, application tasks and services.

`app_main`
System entry point initializing peripherals, mutexes, queues, and pinning all FreeRTOS tasks to Core 0 and Core 1.

`bcd_to_bin`
Converts Binary-Coded Decimal (BCD) values from the RTC registers into standard binary integers.

`bin_to_bcd`
Converts standard integers into Binary-Coded Decimal format for setting RTC time registers.

`rtc_read`
Communicates over I2C to fetch current hours, minutes, and seconds from the DS1307 real-time clock.

`rtc_set`
Updates the DS1307 RTC time registers over I2C when a manual time override command is issued.

`lcd_send_nibble`
Transmits 4-bit command/data nibbles and control bits to the PCF8574 I2C LCD backpack.

`lcd_init`
Initializes the 20x4 LCD display rows, columns, and custom character settings.

`mfrc_read`
Reads an 8-bit internal register from the MFRC522 RFID reader over the shared SPI bus.

`mfrc_write`
Writes data bytes to an MFRC522 RFID reader register over SPI.

`mfrc_transceive`
Sends command frames and receives bit responses during ISO/IEC 14443A card communication.

`rfid_read_uid`
Polls the RFID reader antenna to detect cards, resolve anti-collision, and extract the card UID string.

`servo_write`
Generates PWM duty cycle pulses to mechanically rotate gate and room servos to locked/unlocked angles.

`read_ldr`
Samples analog ambient light levels via the ESP32 ADC channel for automatic room lighting control.

`beep`
Activates the buzzer hardware pin for a specified duration to generate audio feedback.

`send_log`
Formats and pushes audit log events (entry, exit, system actions) safely into the logging queue.

`log_write`
Takes queued log entries and appends them to log.csv on the SPI SD card via VFS FATFS.

`schedule_open_locked`
Evaluates current RTC time against operating hours to determine if the building is officially open.

`is_admin`
Validates whether the current serial console session holds active administrator privileges.

`run_command`
Parses and executes administrator CLI commands (status, login, register, time, etc.).

`access_task`
FreeRTOS core task continuously polling entry and exit RFID readers for access control decisions.

`door_task`
FreeRTOS asynchronous task managing gate servo actuation, hold timers, and auto-closing.

`clock_task`
FreeRTOS background task updating time, managing schedules, and executing the 60-second warning sequence.

`room_task`
FreeRTOS task polling PIR motion and LDR sensors to automate lecture room lighting.

`lcd_task`
FreeRTOS task periodically refreshing the 20x4 display with system status, time, and warning banners.

`logger_task`
FreeRTOS background task draining the log queue and committing records to the SD card.

`cli_task`
FreeRTOS interactive UART task managing terminal input, line buffers, backspace, and command execution.

PART 02

Member Allocation

Five modules of comparable effort, assigned by module ownership.

01

MEMBER 01

System Boot & Time Management Module

`app_main · rtc_read · rtc_set · bcd_to_bin · bin_to_bcd · schedule_open_locked`

System initialization, task scheduling, FreeRTOS core pinning, and real-time clock synchronization.

02

MEMBER 02

RFID Access Control & Security Pipeline

`mfrc_read · mfrc_write · mfrc_transceive · rfid_read_uid · access_task`

SPI bus configuration, dual RFID reader polling, card UID extraction, and access validation logic.

03

MEMBER 03

Actuators, Servos & Room Automation Subsystem

`servo_write · read_ldr · beep · room_task · door_task`

PWM servo gate control, PIR motion sensing, ADC light readings, buzzer alerts, and automated room lighting.

04

MEMBER 04

Display, UI & SD Card Logging Subsystem

`lcd_init · lcd_send_nibble · lcd_task · log_write · send_log · logger_task`

I2C LCD screen rendering, warning banners, SPI SD card mounting, CSV file logging, and audit tracking.

05

MEMBER 05

Serial CLI & Admin Command Processor

`cli_task · run_command · is_admin · clock_task`

UART interactive terminal management, admin authentication, command handling (register, status, time), and safety validation, plus testing the 60-second evacuation lockdown warning.